PH163 scoring validated on ProteinGym DMS (exploratory, 2026-11-05)

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Not pre-registered. Raw results: `experiments/ph163/proteingym_kaggle_validation_2026-11-05.json`, *per assay:* `experiments/ph163/proteingym_per_assay_2026-11-05.csv`.

1 What was tested

Does the PH163 scorer (ESM-2 650M masked-marginal, `scripts/serve/ph163_service.py`, model revision `08e4846e...`) rank real mutational-scanning variants the way the published ESM-2 (650M) control does? Setup: ProteinGym v1.x substitutions (HF revision `1ea2aa10...`, GitHub reference commit `144fe22b...`), run on a free Kaggle Tesla T4 (torch 2.11.0+cu130), 35.3 minutes. 2,465,767 variants in 217 assays; 201 assays scored (16 are longer than 1,022 residues and were skipped, so the numbers below say nothing about long proteins). Metric: Spearman per assay, as in ProteinGym. Control: the official per-assay "ESM2 (650M)" column from the ProteinGym reference table.

2 Result

	mean Spearman (201 assays)
PH163 scorer	0.446
Official ESM-2 (650M)	0.446
Difference	−0.0000, 95% assay-bootstrap CI [−0.0001, 0.0000]

Across assays the two agree at Pearson 1.0, median absolute difference 0.0002 (the official table is rounded to 3 decimals). By selection type PH163 vs official: Activity 0.433/0.433, Binding 0.356/0.356, Expression 0.440/0.440, Organismal fitness 0.396/0.396, Stability 0.523/0.523. Three Arabidopsis assays: 0.699, 0.505, 0.675.

3 What this means, and what it does not

- It means the PH163 implementation is correct: it reproduces the published ESM-2 650M numbers, so scan, percentile and ranking carry no bug of ours. This is a reproduction check, not a new result.
- It does NOT mean PH163 is a good predictor in absolute terms. It inherits ESM-2 650M's accuracy: strong on stability (0.52), weaker on binding (0.36) and viruses (0.29). Larger or MSA/structure-aware models in the same table score higher; we did not run them.
- Only single-site scoring was validated. Multi-site re-scoring, the allowlist screen, ThermoMPNN ddG and ESMFold are validated separately (see the ESMFold and ThermoMPNN reports).

- 16 assays (> 1,022 residues) skipped; the production window logic for long proteins was therefore not checked against DMS data here.